

Definition and the Fundamental

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1 Definitions

Target: claims like

- (1a) To be a vixen is to be a female fox.
- (1b) To be just is to have a well-ordered soul.
- (1c) To be a hydrogen atom is to be an atom with a nucleus containing exactly one proton.
- (1d) For two lines to be parallel is for there to be a line that is perpendicular to both of them.

Readings I'm not concerned with:

- (2a) To enter the world of Hayao Miyazaki is to experience a kind of lighthearted enchantment that is unique to the world of animation. (R. Ebert)

- (2b) To be a vixen is to be a fox.

I'm interested in a reading of 'to ϕ is to ψ ' on which this claim entails that—and explains why—it's necessary that something ϕ s iff it ψ s.

- Are definitions just property identities? Not exactly: clauses like 'to be a vixen' [or 'PRO to be a vixen'] are not syntactically interchangeable with paradigmatic "singular terms":
 - (3) He asked me to water his plants
 - *He asked me the property of watering his plants.

2 Notation

A binary operator that binds zero or more variables:

x is a vixen $:=_x$ x is a female fox

x is parallel to y $:=_{xy}$ $\exists z(z$ is perpendicular to $x \wedge z$ is perpendicular to $y)$

There are vixens $:=$ there are female foxes

Not all the variables need to be bound by $:=$, so we can say things like

$\exists y(y \text{ is a country} \wedge (x \text{ is German} :=_x x \text{ is from } y))$

Use $\phi ::= \psi$ as shorthand for $\phi :=_{v_1 \dots v_n} \psi$, where $v_1, \dots, v_n$ are all the variables free in either ϕ or ψ .

*Logic:*¹

- (Truth-preservation) $\phi :=_{v_1 \dots v_n} \psi, \phi \vdash \psi$
- (Reflexivity) $\vdash \phi :=_{v_1 \dots v_n} \phi$
- (Transitivity) $\phi :=_{v_1 \dots v_n} \psi, \psi :=_{v_1 \dots v_n} \chi \vdash \phi :=_{v_1 \dots v_n} \chi$
- (Symmetry) $\phi :=_{v_1 \dots v_n} \psi \vdash \psi :=_{v_1 \dots v_n} \phi$

Connective $\circ$ (taking m wffs to make a wff) is *transparent* iff

$\phi_1 :=_{v_1 \dots v_n} \psi_1, \dots, \phi_m :=_{v_1 \dots v_n} \psi_m \vdash \circ(\phi_1, \dots, \phi_m) :=_{v_1 \dots v_n} \circ(\psi_1, \dots, \psi_m)$.²

3 My questions

To what extent can we use this ideology to illuminate debates about *fundamentalness* (primitiveness, priority, grounding, reduction...)? Is there anything properly metaphysical at stake in such debates beyond what shows up in the pattern of definitions one accepts and rejects?

- Not all such debates deserve to be made sense of. We should want *sometimes* to be able to say that it's a matter of indifference whether one regards X as primitive and Y as defined, or vice versa. For example, it isn't attractive to think of one side of ' x loves y $:= y$ is loved by x ' as "more fundamental" than the

¹Also need axioms for (a) performing uniform substitutions of variables across a whole definition; (b) subtracting variables from the list of those bound by $:=$; (c) adding variables to the list, at least when they are not free on either side.

²More generally, if $\circ$ takes m wffs and binds k variables, it is transparent iff

$$\phi_1 :=_{v_1 \dots v_n} \psi_1, \dots, \phi_m :=_{v_1 \dots v_n} \psi_m \vdash \circ_{v_1, \dots, v_k}(\phi_1, \dots, \phi_m) :=_{v_{k+1} \dots v_n} \circ_{v_1, \dots, v_k}(\psi_1, \dots, \psi_m).$$

other.

Hope: show how *some* debates about fundamentalness can be understood as debates about which definitions are true, *modulo* certain background theories about how definitions hang together.

4 Motivating the background theories

Given that definitions are supposed to explain necessities...

- (i) There is something wrong with “giving multiple definitions of the same thing”. I can’t say both ‘for x to be parallel to y is for x and y to be lines that don’t intersect’ and ‘for x to be parallel to y is for there to be a z that is perpendicular to both’, and then claim to have explained the necessity of
 - (*) Whenever z is perpendicular to two lines x and y , x and y don’t intersect.
- (ii) There is something wrong with “circular definitions”. I can’t say ‘For x to be perpendicular to y is for x to be perpendicular to y and to no other line that intersects y ’ and thereby claim to have explained the necessity of (*).
- (iii) There is something wrong with “infinitely regressive definitions”. I can’t endorse the infinite series of definitions ‘For x to be perpendicular _{n} to y is for x to be perpendicular _{$n+1$} to y and to no other line that intersects y ’ and thereby claim to have explained the necessity of (*).

But what could this mean, given that we can have (i) $\phi ::= \psi_1$ and $\phi ::= \psi_2$, where ψ_1 and ψ_2 are distinct formulae, and (ii) we can (and indeed always do) have $\phi ::= \phi$, and (iii) we can have $\phi_n ::= \phi_{n+1}$ for some infinite set of formulae?

5 The demand for a minimal definitional basis

First thought: “The direction of a definition is from the syntactically simple to the syntactically complex.”

- (i) $\chi ::= \bigcirc(\phi_1, \dots, \phi_m)$ and $\chi ::= \bigcirc(\psi_1, \dots, \psi_m)$ can both be true only if $\bigcirc = \odot$ and $\phi_1 ::= \psi_1, \dots, \phi_n ::= \psi_n$ are all true.

- (ii) $\phi(v_1, \dots, v_n) ::= \bigcirc(\psi_1, \dots, \psi_k, \phi(u_1, \dots, u_n), \psi_{k+1}, \psi_m)$ is never true.
- (iii) We never have an infinite chain of true definitions in which syntactic complexity increases without bound.

The first thought is bad. For any specifiable function f from formulas to formulas (e.g. the identity function), we are free to introduce a connective $\bigcirc$ by stipulating that $\ulcorner \bigcirc(\phi) \urcorner$ is synonymous with $f(\phi)$.

Second thought: (i)–(iii) are true when we restrict the connectives involved to members of some special set of connectives (presumably all transparent), and understand ‘syntactic complexity’ in terms of depth of embedding under these connectives.³

A different way to put the thought: Let $\mathcal{L}$ be an interpreted language containing $::=$, and $\mathcal{L}^-$ a sublanguage of $\mathcal{L}$ all of whose connectives are transparent in $\mathcal{L}$.⁴ ($\mathcal{L}^-$ may have fewer connectives than $\mathcal{L}$, and may feature as atoms some things which count as syntactically complex in $\mathcal{L}$).

$\mathcal{L}^-$ is a **definitional basis** for $\mathcal{L}$ iff for every formula ϕ of $\mathcal{L}$, there is a formula ψ of $\mathcal{L}^-$ such that no variable occurs free in ψ but not ϕ , and $\phi ::=_{v_1 \dots v_n} \psi$ is true in $\mathcal{L}$ under some assignment (where $v_1, \dots, v_n$ are the variables free in ϕ).

$\mathcal{L}^-$ is a **minimal definitional basis** for $\mathcal{L}$ iff $\mathcal{L}^-$ is a definitional basis for $\mathcal{L}$, and there are no two formulae $\phi(v_1, \dots, v_n), \psi$ of $\mathcal{L}^-$ for which $\phi ::=_{v_1 \dots v_n} \psi$ is true in $\mathcal{L}$ under any assignment (except when ϕ and ψ differ merely by substitution of variables).

The claim that (i)–(iii) are true when restricted to a given set of transparent connectives is (more or less) equivalent to the claim that there

³Given the transparency of the special connectives, (ii) is redundant, since it follows from (iii).

⁴It wouldn’t matter much if we dropped this requirement. Given an $\mathcal{L}^-$ that was a minimal definitional basis for $\mathcal{L}$ with non-transparent connectives, we can define a new minimal definitional basis $\mathcal{L}^*$ with only transparent connectives by substituting for each connective $\bigcirc$ of $\mathcal{L}$ a new connective $\odot$ stipulatively defined by $\odot(\phi_1 \dots \phi_n) ::=_{v_1 \dots v_n} \bigcirc(\psi_1 \dots \psi_n)$, where ψ_i is the unique formula of $\mathcal{L}^-$ for which $\phi_i ::=_{v_1 \dots v_n} \psi_i$.

is a minimal definitional basis $\mathcal{L}^-$ with just those connectives.⁵

6 Weakening the demand

Reasons to worry that the demand is too strong:

- The need to choose between $\phi \vee \psi$ and $\psi \vee \phi$ seems to be an artifact of our linear language, no more significant from the metaphysical point of view than the choice of a typeface or a pitch. But we can't have $(\phi \vee \psi) ::= (\psi \vee \phi)$ if there's a minimal basis that includes $\vee$.
- 'We might, for instance, express negation not by writing a word "not", but by writing what we negate upside-down. Such a symbolism is only inconvenient because we are not trained to perceive complicated symmetry about a horizontal axis, and if we adopted it we should be rid of the redundant "not-not", for the result of negating the sentence "p" twice would simply be the sentence "p" itself.' (Ramsey, 'Facts and Propositions')
Looks like a good reason to want $\phi ::= \neg\neg\phi$.⁶ But we can't have this if there's a minimal basis that includes $\neg$.
- Q: How far do these thoughts generalise? Is there a principled way to draw the line?

R-minimality

Let $\mathcal{L}^-$ be a definitional basis for $\mathcal{L}$, and let R express a consequence relation on formulae of $\mathcal{L}^-$. Say that $\mathcal{L}^-$ is *R-minimal* iff for any formulae $\phi, \psi(v_1, \dots, v_n)$ of $\mathcal{L}^-$, $\phi :=_{v_1 \dots v_n} \psi$ is true in $\mathcal{L}$ (on any assignment) *only if* $\phi \dashv\vdash_R \psi$.

⁵*Sketch of proof:* Say that a formula ϕ with free variables $x_1, \dots, x_n$ is *impure* if there is a formula ψ in which one or more variables $y_1, \dots, y_m$ occur free that are not free in ϕ , such that $\phi :=_{x_1 \dots x_n y_1 \dots y_m} \psi$ is false but $\phi :=_{x_1 \dots x_n} \psi$ is true under at least one assignment. Otherwise, ϕ is *pure*. For the atomic formulae of $\mathcal{L}^-$, choose one member from each $:=$ -equivalence class of pure $\mathcal{L}$ -formulae which contains no member whose main connective belongs to the given set. By (iii) and the transparency of the connective, every pure $\mathcal{L}$ -formula is $:=$ -equivalent to some $\mathcal{L}^-$ formula; hence every impure $\mathcal{L}$ -formula is $:=$ -equivalent to some $\mathcal{L}^-$ formula under some assignment. The construction guarantees that no atomic $\mathcal{L}^-$ -formula is $:=$ -equivalent to any other $\mathcal{L}^-$ -formula; and by (i), no two syntactically complex $\mathcal{L}^-$ -formulae are $:=$ -equivalent.

⁶And by the same token $\neg(\phi \vee \psi) ::= (\neg\phi \wedge \neg\psi)$, $\neg\exists x\phi ::= \forall x\neg\phi$, $\phi \supset \psi ::= \neg\phi \vee \psi$, $\phi \vee \psi ::= \neg\phi \supset \psi$.

7 Minimal bases and fundamentality

Idea: a vocabulary is a "candidate to be fundamental" iff it is included in a [R-]minimal definitional basis [meeting such-and-such further conditions. . .].

Too many bases?

Suppose that to be *grue* is to be either green and early or blue and not early. Consider two sublanguages:

$\mathcal{L}^-$: predicates 'Green', 'Blue', 'Early'; connectives $\wedge, \vee, \neg, \forall, \exists$.

$\mathcal{L}^*$: predicates 'Grue', 'Bleen', 'Early'; connectives $\underline{\wedge}, \underline{\vee}, \underline{\neg}, \underline{\forall}, \underline{\exists}$.

Since every formula in each of these sublanguages is equivalent (in first-order predicate calculus) to infinitely many others, it is easy to define a bijection f from formulas of $\mathcal{L}^*$ to formulas of $\mathcal{L}^-$ such that

$f(\ulcorner \text{Grue}(v) \urcorner) = \ulcorner \text{Green}(v) \wedge \text{Early}(v) \vee (\text{Blue}(v) \wedge \neg \text{Early}(v)) \urcorner$,
and similarly for 'Bleen'.

$f(\ulcorner (\text{Grue}(v) \underline{\wedge} \text{Early}(v)) \underline{\vee} (\text{Bleen}(v) \underline{\wedge} \underline{\neg} \text{Early}(v)) \urcorner) = \ulcorner \text{Green}(v) \urcorner$,
and similarly for 'Blue'.

$f(\ulcorner \phi \underline{\wedge} \psi \urcorner)$ is always logically equivalent to $\ulcorner f(\phi) \wedge f(\psi) \urcorner$, and similarly for the other connectives.

Define the underlined connectives in such a way that for any formula ϕ of $\mathcal{L}^*$, $\ulcorner \phi \urcorner ::= f(\ulcorner \phi \urcorner)$ is true. Then if $\mathcal{L}^-$ is a minimal basis, $\mathcal{L}^*$ is too.

Responses:

- We could try to understand questions about fundamentality in a way that somehow builds in a prejudice in favour of $\wedge, \vee$, etc. over their underlined counterparts. Perhaps this isn't bad—after all it's not as if we could have hoped to explain 'fundamental' without using any logical vocabulary! But clearly the current framework does less than we might have hoped to illuminate debates about which connectives are fundamental.
- Q: Should the possibility of these constructions make us more hospitable to the idea that logical equivalents are fully inter-substitutable in the scope of $:=$?

8 Consolation?

Suppose we have two definitional bases, $\mathcal{L}^-$, with connectives $\wedge, \vee, \neg, \forall, \exists$, and $\mathcal{L}^*$, with connectives $\underline{\wedge}, \underline{\vee}, \underline{\neg}, \underline{\forall}, \underline{\exists}$, such that $\mathcal{L}^-$ is R -minimal and $\mathcal{L}^*$ is R^* -minimal, where R and R^* are the usual first-order consequence relations on $\mathcal{L}^-$ and $\mathcal{L}^*$. And suppose that these consequence relations are *compatible*, in the sense that for any formulae ϕ_1, ϕ_2 of $\mathcal{L}^-$ and ψ_1, ψ_2 of $\mathcal{L}^*$ such that $\phi_1 ::= \psi_1$ and $\phi_2 ::= \psi_2$, $\psi_1 \vdash_{R^*} \psi_2$ iff $\phi_1 \vdash_R \phi_2$. Then:

Result A The logical vocabularies of $\mathcal{L}^-$ and $\mathcal{L}^*$ are equivalent, in the sense that, for example

($\wedge$) Whenever $\phi_1 ::= \psi_1$, $\phi_2 ::= \psi_2$ and $(\phi_1 \wedge \phi_2) ::= \psi_3$,
 $(\psi_1 \underline{\wedge} \psi_2) \dashv\vdash_{R^*} \psi_3$.

($\forall$) Whenever $\phi ::= \psi_1$ and $\forall v_1 \phi ::= \psi_2$, $\underline{\forall} v_1 \psi_1 \dashv\vdash_{R^*} \psi_2$.

Result B $\mathcal{L}^-$ and $\mathcal{L}^*$ have the same number of n -place atomic predicates, for each n .

Idea of proof of Result A:

($\wedge$) By the ‘compatibility’ of $\vdash_R$ and $\vdash_{R^*}$, $\psi_3 \vdash_{R^*} \psi_1$ and $\psi_3 \vdash_{R^*} \psi_2$, hence $\psi_3 \vdash_{R^*} \psi_1 \underline{\wedge} \psi_2$. Conversely: whenever $\phi_4 \vdash_R \phi_1$ and $\phi_4 \vdash_R \phi_2$, $\phi_4 \vdash_R \phi_1 \wedge \phi_2$; so whenever $\psi_4 \vdash_{R^*} \psi_1$ and $\psi_4 \vdash_{R^*} \psi_2$, $\psi_4 \vdash_{R^*} \psi_3$, so in particular $\psi_1 \underline{\wedge} \psi_2 \vdash_{R^*} \psi_3$.

($\forall$) By compatibility, $\psi_2 \vdash_{R^*} \psi_1$. Since v_1 is not free in ψ_2 , $\psi_2 \vdash_{R^*} \underline{\forall} v_1 \psi_1$, since, since $\forall v_1 \phi \vdash_R \phi$. Conversely: whenever $\phi' \vdash_R \phi$ and doesn’t have v_1 free, $\phi' \vdash_R \forall v_1 \phi$; so whenever $\psi' \vdash_{R^*} \psi_1$ and doesn’t have v_1 free, $\psi' \vdash_{R^*} \psi_2$; so in particular $\underline{\forall} v_1 \psi_1 \vdash_{R^*} \psi_2$.

Idea of proof of Result B: Say that a first-order formula ϕ is *n-maximal* iff (i) the free variables in ϕ are exactly $x_1 \dots x_n$; (ii) ϕ is logically consistent; (iii) for any other formula ψ with at most $x_1 \dots x_n$ free, either ϕ logically entails ψ or ϕ is logically inconsistent with ψ ; (iv) ϕ entails $x_i \neq x_j$ for all $i < j \leq n$; (v) ϕ entails $\forall y(y = x_1 \vee \dots \vee y = x_n)$, and (vi) ϕ entails $\exists y(y = x_i)$ for each $i \leq n$. In a first-order language with k_0 sentence-letters, k_1 1-place predicates, k_2 2-place predicates (not including $=$), $\dots$, k_m m -place predicates, the number of equivalence

classes of n -maximal formulas under logical equivalence is

$$2 \uparrow (k_0 + nk_1 + n^2k_2 + \dots + n^mk_m)$$

By **Result A**, if a formula ϕ of $\mathcal{L}^-$ is n -maximal, and $\phi :=_{x_1 \dots x_n} \psi$ for some formula ψ of $\mathcal{L}^*$, then ψ is n -maximal too. So (using ‘compatibility’), the number of equivalence classes of n -maximal formulae is the same in $\mathcal{L}^-$ and $\mathcal{L}^*$. So if $\mathcal{L}^-$ has k_0 0-ary predicates. . . and k_m m -ary predicates, and $\mathcal{L}^*$ has j_0 0-ary predicates. . . and j_m m -ary predicates, then for every n ,

$$2 \uparrow (k_0 + nk_1 + n^2k_2 + \dots + n^mk_m) = 2 \uparrow (j_0 + nj_1 + n^2j_2 + \dots + n^mj_m)$$

This is an infinite set of linearly independent equations in finitely many unknowns, so it has a unique solution.

Extensions:

- Things still work if we take $\vdash_R$ to be any kind of free logical consequence.⁷ This is significant: it’s implausible that $\forall y \exists x(x = y) := \forall x(x = x)$.
- We can allow $\vdash_R$ to be a bit stronger than first-order consequence, e.g. by allowing certain predicates to count as symmetric as a matter of logic. Then we can strengthen ($\forall$) to claim that $\mathcal{L}^-$ and $\mathcal{L}^*$ have the same number of symmetric 2-place predicates ($Rxy :=_{xy} Ryx$), the same number of totally symmetric 3-place predicates, the same number of 3-place predicates that are symmetric in two of their argument places, the same number of 3-place predicates that are “rotationally” symmetric ($Rxyz :=_{xyz} Ryzx$), etc.
- We can expand the syntax of the languages in various ways, e.g. by allowing plural (or higher-order) quantification. Then the relevant version of **Result B** will also require that $\mathcal{L}^-$ and $\mathcal{L}^*$ have the same number of predicates taking one plural argument, the same number of predicates taking one plural and one singular argument, etc.

⁷For ($\forall$), use the fact that in free logic, if x isn’t free in ϕ and $\phi \wedge \psi \vdash_R x = y$, $\psi \vdash_R \exists y(x = y)$.